

Synthesis, Characterisation And Antibacterial Activity Of Some New Schiff Bases.

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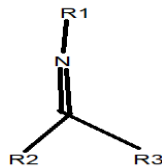
Abstract-

Schiff's bases are compounds of the aldehyde or ketone type in which the carbonyl group is replaced by an imine or azomethine group. It contains a carbon-nitrogen double bond with the nitrogen atom connected to an aryl or alkyl group but no hydrogen. Many metal complexes of Schiff bases have been studied to treat various bacterial, fungal, viral infections and neurological disorders. Metal complexes have a high place in pharmaceutical chemistry due to their different reactions to non-metals. Extensions of biological activity from the implementation of transition metals at Schiff bases have been reported. Schiff bases are widely used in analytical, organic and inorganic chemistry. Among azomethane derivatives, C = N connections are various natural (antistrocladidine has antimalarial activity), natural-derived (chitosin-derived shift bases have antifungal action) and non-compatible compounds required for biological activity. Many azomethanes have important antibacterial, antifungal, anticancer and diuretic activities. In addition to biological activities, finds application in many other fields such as organic synthesis, dyes, pigments, polymer stabilizers, corrosion inhibitors, fungicides, agrochemicals, analytical chemistry, electrical conductivity, magnetism, host guest chemistry and catalysis. Due to these important applications, we have synthesized various new Schiff bases and other heterocyclic compounds along with their characterisation (IR, ¹HNMR and mass) and antibacterial evaluation on plant and human pathogenic bacterial strains. All the compounds showed excellent antibacterial activity against the pathogenic strains. **Keywords:** Substituted amines, Schiff bases, antibacterial activity, plant pathogens,

INTRODUCTION

Schiff's bases have received a lot of attention in recent years due to their wide use as dyes and pigments, catalysts, intermediates in organic synthesis, polymer stabilizers and due to their wide range of biological activities, including antifungal, antibacterial, antimalarial, anti-proliferative, anti-inflammatory, antiviral and antipyretic properties¹⁻⁴. Schiff's bases are compounds of the aldehyde

or ketone type in which the carbonyl group is replaced by an imine or azomethine group. It contains a carbon-nitrogen double bond with the nitrogen atom connected to an aryl or alkyl group but no hydrogen.



The structure of a Schiff base

Schiff's bases are of the general formula $R_1N=CR_2R_3$, where R_1 is a phenyl or alkyl group which gives the Schiff base its stability⁵. Although the formation of Schiff's bases is reversible, due to hydrolysis of imine under certain conditions, it always is, simple for the reaction to succeed. It is still not clear which type of Schiff's base is stable in the presence of water even with an acidic solution, while others are very sensitive to water and readily hydrolyse to aldehyde. To overcome this potential hydrolysis, the Schiff base reaction must be carried out under dry solvent conditions or using an additional procedure to remove the side product, water, in the imine formation. The lone pair on the nitrogen atom of the imine moiety can provide electrons which allow the formation of a suitable donor bond to a metal ion for complexation to occur. Many Schiff bases have a second functional group, normally OH and SH groups or another N atom, which are close to the imine group. These functional groups allow the formation of five or six chelate rings when coordinated with metal ions. Many metal complexes have been studied to treat various bacterial, antifungal (Schiff base - bismuth complex), antiviral (Schiff base - Silver complex) various neurological disorders and anticancer drugs. Metal complexes have a high place in pharmaceutical chemistry due to their different reactions to non-metals. Studies have shown that metal complexes show more biological activity than free organic compounds. Extensions of biological activity from the implementation of transition metals at Schiff bases have been reported. In fact, lipophilization of a drug is enhanced by the formation of chelates that enhance the action of the drug due to its effective permeability at the site of action⁶.

Schiff bases are widely used organic compounds that bind to metal ions by azomethane nitrogen and are widely used in many fields, including analytical, organic and inorganic chemistry. Among azomethane derivatives, C = N connections are various natural (antistrocladidine has antimalarial activity), natural-derived (chitosin-derived Schiff bases have antifungal action) and non-compatible compounds required for biological activity. Many azomethanes have important antibacterial, antifungal, anticancer and diuretic activities. A large number of Schiff-based complexes have potential biological interests that are being used as successful models of biological compounds. In addition to biological activities, he has been involved in many fields such as organic synthesis, dyes, pigments, polymer stabilizers, corrosion inhibitors, fungicides, agrochemicals, analytical chemistry, electrical conductivity, magnetism, host guest chemistry, ion exchange, and nonlinearity optics and catalysts⁷.

Schiff base complexes play a major role in homogeneous and heterogeneous catalysis. They have been shown to be catalytic in the carbonization of alcohols and alkanes at low pressure, polymerization of olefins, oxidation of hydrocarbons, reduction of ketones to alcohols, the Heck reaction, isomerization of norbornadiene to quadricyclane etc⁸⁻¹¹. Cycloalkenes ring opening is possible with transition metals in the presence of alkylating agents at high temperatures without any control over the molecular weight of the polymer¹². On the other hand, when catalysed by the Schiff base complex, it occurs with considerable control over the molecular weight of the polymer at low temperatures without any reaction. Phosphine Schiff base complex enhances enantioselectivities in hydrosilylation reactions¹³. Due to these important applications, there is also a lot of interest in the synthesis of Schiff base complexes.

REVIEW OF LITERATURE

Extensive literature surveys have been conducted on the synthesis and biological activity of Schiff bases. Literary surveys have shown that the Schiff base exhibits a variety of biological actions based on azomethane linkage, which causes various antibacterial, antifungal and antioxidant properties. Several studies have shown that the presence of a single pair of electrons in the sp^2 hybridized orbital of the nitrogen atom of the azomethane group is of significant chemical and biological importance¹⁴. They interfere with normal cell processes by forming hydrogen bonds between the active centers of the cell components and the sp^2 hybridized nitrogen molecules of the azomethine group. Considering all these facts and the effectiveness of Schiff base derivatives, the novel Schiff base derivatives are synthesized and tested for antimicrobial activity. S. Malladi *et al*¹⁵ synthesized Pyrazole-based Schiff bases, 3 (a-j), 4 - [(3-substitute-1H pyrazol-3-yl) methyleneamino]-5-substituted-4H-1,2,4-triazole-3-thiols, and screened for antibacterial activity against *S. aureus*, *P. aeruginosa*, *B. subtilis* and *E. coli*. From the results, the compound is 3f found equipotent against *P. aeruginosa*, *B. subtilis* and *E. coli* and highly active against *S. aureus* compare to standard drug ceftriaxone. K. V. Sashidhara *et al*¹⁶ synthesized a series of novel keto-enamine Schiff bases, derived from 8-hydroxy-quinoline and subjected for *in vitro* antioxidant, *in vivo* antidyslipidemic, and postheparin lipolytic activity. Lei Shi *et al*¹⁷ had reported the synthesis and antimicrobial activity of a series of Schiff bases derived from the condensation of 5-chloro-salicylaldehyde and primary amines has recently been reported. Avinash Shinde *et al*¹⁸ had synthesized a new class of bis-Schiff bases by the reaction between propane-1,3-diamine with halogen substituted benzaldehydes by conventional as well as microwave irradiation technique. Kalpana V. Sontakke *et al*¹⁹ has reported synthesis and antimicrobial activity of some new bis-Schiff bases. O-phenylene diamine (II) was condensed with substituted salicylaldehyde or substituted O-hydroxy acetophenone by refluxing in methanol and traces of acetic acid to get desired bis Schiff bases. Karamunge K. G. *et al*²⁰ had reported synthesis and antimicrobial activity of some new Schiff bases. New Schiff bases were synthesized by refluxing the reaction mixture of 4-iodoaniline, 4-chloroaniline, and p-toluidine with halogenohydroxy substituted acetophenones in methanol in the presence of glacial acetic acid. Subhash B. Junne *et al*²¹ in continuation to their earlier research work had reported for first time a typical condensation

between substituted aryl amines and indole-3-aldehyde at room temperature with combination of grinding to afford heterocyclic imines. The antibacterial activities of the synthesized compounds were determined by agar well diffusion method. Archana Y. Vibhute *et al*²² had reported an environmentally benign organic solvent free approach for synthesis of new Schiff bases and evaluation of antibacterial activity. Subhash B Junne *et al*²³ had carried out synthesis and antimicrobial activity of some new Schiff bases and 2-Azetidinones containing iodo hydroxy biphenyl moiety. All synthesized compounds were assessed for their antibacterial. Santosh *et al*²⁴ synthesized Schiff bases (sulfonamide), which was prepared by the reaction of 4-aminobenzenesulphonamide and substituted aromatic aldehydes. Aqeel *et al*²⁵ synthesized three new series of biologically active (against bacteria, fungi, yeast etc.) amino substituted Schiff bases. Zarei *et al*²⁶ synthesized azoaldehyde from *p*-anisidine and 2-hydroxy-3-methoxy benzaldehyde (*o*-vanillin) in *aqueous* medium at 0-5 °C.

In present investigation Schiff bases were synthesized by condensation of substituted aromatic amines with substituted aromatic aldehydes. This method has been implemented for the synthesis of Schiff bases because under moderate reaction conditions it yields unambiguous products with excellent yield.

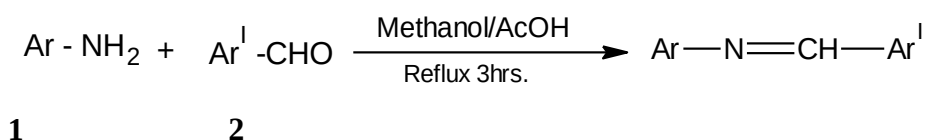
PRESENT WORK AND METHOD

The chemistry of the Schiff bases is still a flourishing field. The main feature of Schiff bases is that they act as starting materials for the synthesis of heterocyclic compounds such as 4- thiazolidinone. The utility lies in their usefulness as a synthon. Schiff bases have a large number of useful biological activities such as antibacterial, antifungal, pesticides, antifungaetc.

General Method for Synthesis of Substituted Schiff Bases

Equimolar quantity of substituted aromatic amines and substituted aromatic aldehydes in requisite methanol containing few drops of acetic acid was refluxed for 3 hours. It was allowed to cool, poured with stirring in cold water. The resulting solid was filtered, washed with cold water and recrystallised from ethanol. The purity of synthesized Schiff bases was checked by T.L.C. The structures of the Schiff bases were assigned on the basis of spectral data. (IR, ¹HNMR, and Mass spectral analysis).

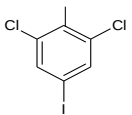
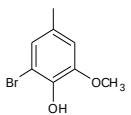
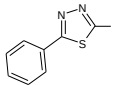
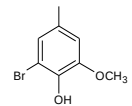
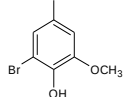
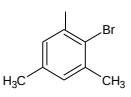
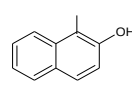
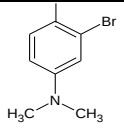
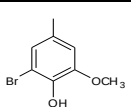
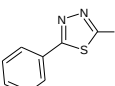
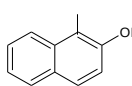
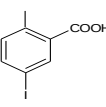
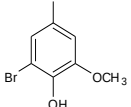
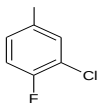
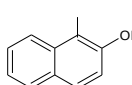
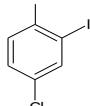
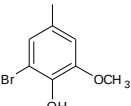
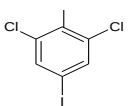
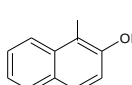
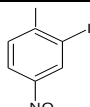
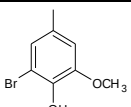
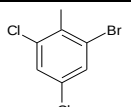
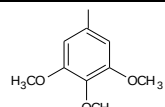
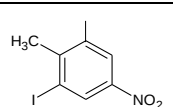
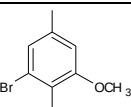
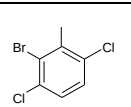
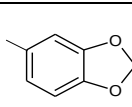
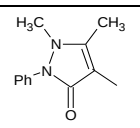
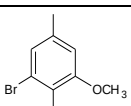
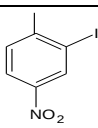
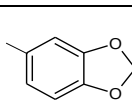
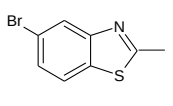
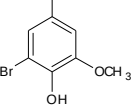
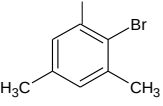
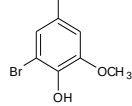
Scheme-I



I(a-r)

TABLE 1. Scheme- (a-r)

Comp. Code	Ar	Ar ^I	Comp. Code	Ar	Ar ^I

Ia			Ij		
Ib			Ik		
Ic			Il		
Id			Im		
Ie			In		
If			Io		
Ig			Ip		
Ih			Iq		
Ii			Ir		

EXPERIMENTAL

1. Preparation of 4-[(1*E*)-(3-bromo-4-hydroxy-5-methoxyphenyl) methylidene] amino}-1, 5-dimethyl-2-phenyl-1, 2-dihydro-3*H*-pyrazol-3-one (Ih)

A mixture of 3-bromo-4-hydroxy-5-methoxyphenyl) (0.01 mmoles) and 4-amino-1, 5-dimethyl-2-phenyl-1, 2-dihydro-3*H*-pyrazol-3-one moles (0.01 mmoles) was dissolved in methanol (20 mL). Few drops of acetic acid was added and refluxed for 3 hrs. The resultant solution was cooled and poured in cold water. The separated solid was filtered, recrystallized from ethanol to give Ih.

2. Preparation of 2-bromo-4-[(*E*)-[(2-bromo-3,5-dimethylphenyl)imino]methyl]-6-methoxyphenol (Ir)

A mixture of 3-bromo-4-hydroxy-5-methoxyphenyl) (0.01 mmol) and 2-bromo-3,5-dimethyl aniline moles (0.01 mmol) was dissolved in methanol (20 mL). Few drops of acetic acid was added and refluxed for 3 hrs. The resultant solution was cooled and poured in cold water. The separated solid was filtered, recrystallized from ethanol to give II a.

3. Preparation of 2-bromo-4-((E)-[(2-iodo-4-nitrophenyl)imino]methyl)-6-methoxyphenol (If)

A mixture of 3-bromo-4-hydroxy-5-methoxyphenyl) (0.01 mmol) and 2-iodo-4-nitro aniline moles (0.01 mmol) was dissolved in methanol (20 mL). Few drops of acetic acid was added and refluxed for 3 hrs. The resultant solution was cooled and poured in cold water. The separated solid was filtered, recrystallized from ethanol to give If.

4. Preparation of 2-bromo-4-((E)-[(5-bromo-1,3-benzothiazol-2-yl)imino]methyl)-6-methoxyphenol (Ii)

A mixture of 3-bromo-4-hydroxy-5-methoxyphenyl) (0.01 mmol) and 5-bromo-1,3,5-benzothiazole amine moles (0.01 mmol) was dissolved in methanol (20 mL). Few drops of acetic acid was added and refluxed for 3 hrs. The resultant solution was cooled and poured in cold water. The separated solid was filtered, recrystallized from ethanol to give Ii.

5. Preparation of 2-bromo-6-methoxy-4-((E)-[(5-phenyl-1,3,4-thiadiazol-2-yl)imino]methyl)phenol (Ij)

A mixture of 3-bromo-4-hydroxy-5-methoxyphenyl) (0.01 mmol) and 5-phenyl-1,3,4-thiadiazol-2-amine moles (0.01 mmol) was dissolved in methanol (20 mL). Few drops of acetic acid was added and refluxed for 3 hrs. The resultant solution was cooled and poured in cold water. The separated solid was filtered, recrystallized from ethanol to give Ij.

BIOLOGICAL EVALUATION: PRESENT WORK AND METHOD

Antibacterial Activity²⁷⁻³¹

Despite lot of advances made in drug research, microbial infections still remains one of the major health problems today. The global increase in resistance to antimicrobial drugs has also created several public health issues. The resistance development is a problem whenever antimicrobial drugs are used for the treatment of infections. In recent years, antibiotic-resistant microbial infections have become a growing concern. With the threat of traditional antibiotics and antimicrobial agents becoming obsolete, the development of new classes of antimicrobial agents has become an important subject of current research. This information prompted us to synthesize some new Schiff bases and other heterocyclic compounds with some excellent antibacterial activity.

Pathogenic strain of bacteria which are used in the screening of newly synthesized compounds for their antibacterial activity are as below.

1. *Xanthomonas citri* (Xc)

Xanthomonas axonopodis pv. *citric* is a gram negative, aerobic, rod-shaped virus that causes citrus canker. This pathogen often causes leaf-spotting and fruit disease leading to serious economic losses due to depletion, declining yields, and declining marketing of mature, symptomatic fruit.

2. *Erwinia carotovora* (Ec)

This bacterium is a plant pathogen associated with mild rot of many vegetables. It produces these pathogenic enzymes that hydrolyze pectin between plant cells and cause a disease.

3. *Escherichia coli* (E. coli)

E. coli is a Gram-negative organism that is not contagious. The intestinal *E. coli* can cause many intestinal and extra intestinal infections such as urinary tract infections and mastitis. However, *E. coli* is not always harmful to the human body or other animals. *E. coli* live in our intestine, where they help our body to break down the food we eat, as well as waste processing, vitamin K production and food absorption.

4. *Bacillus subtilis* (Bs)

Bacillus subtilis cells are rod-shaped, Gram-positive bacteria found naturally in soil and vegetation. *Bacillus subtilis* grow in the mesophilic temperature range i.e. 25-35°C. Stress and starvation are common in this environment; therefore, *Bacillus subtilis* has evolved a set of strategies that allow survival under these harsh conditions. One strategy, for example, is the formation of stress-resistant endospores.

Experimental

Antibacterial activities of Synthesized substituted Schiff Bases I (a-r)

The antibacterial activity of synthesized compounds was determined by screening them against gram negative and gram positive culture viz. *Xanthomonas citri*, *Erwinia carotovora*, *Escherichia coli*, *Bacillus subtilis* using cup plate agar diffusion method. The basic principle of antimicrobial assay lies in the comparison of inhibition of growth of microorganism produced by the known concentration of antimicrobial agents to be tested with that produced by known concentration of standard antibacterial agent having known activity.

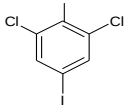
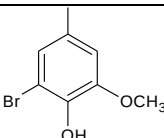
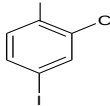
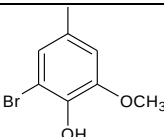
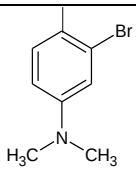
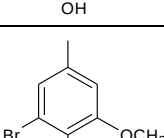
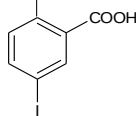
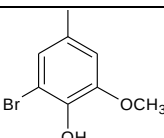
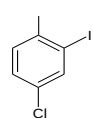
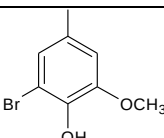
TABLE 2. Materials and Method for antibacterial activity

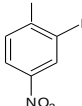
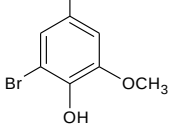
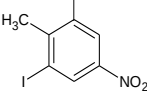
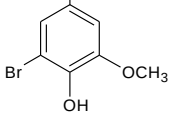
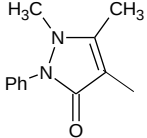
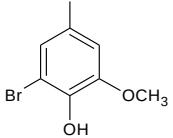
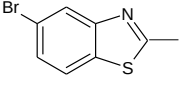
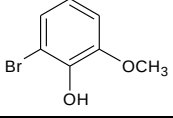
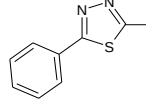
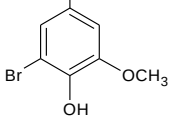
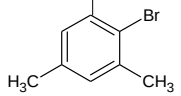
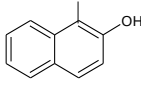
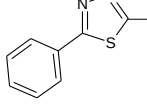
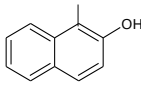
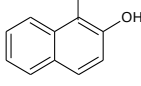
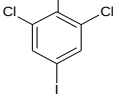
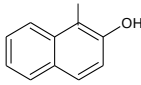
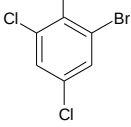
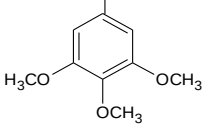
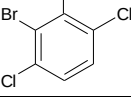
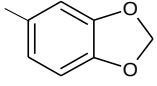
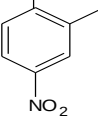
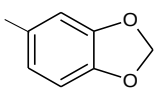
Method	Cup Plate Agar Diffusion Method ³²
Medium	Nutrient agar
Organism	Gram negative and Gram positive <i>Xanthomonas citri</i> , <i>Erwinia carotovora</i> ,

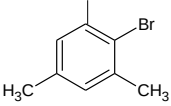
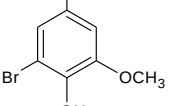
employed	<i>Escherichia coli</i> , <i>Bacillus subtilis</i>
Stock solution	Prepared in Dimethyl formamide for both synthesized compounds and standard compound
Incubation time	24 hrs.
Incubation temp.	37°C
Cup size	6 mm
Control	5%DMF

RESULTS

Table 3. analytical data of synthesized compounds

compounds	Ar	Ar ¹	Molecular Formula	Molecular Weight.	Melting point	Yield
Ia			C ₁₄ H ₉ BrCl ₂ INO ₂	500.94	80°C	92%
Ib			C ₁₄ H ₁₀ BrClINO ₂	466.49	144°C	90%
Ic			C ₁₆ H ₁₆ Br ₂ N ₂ O ₂	428.11	160°C	90%
Id			C ₁₅ H ₁₁ BrINO ₄	476.06	137°C	75%
Ie			C ₁₄ H ₁₀ BrClINO ₂	466.49	119°C	80%

If			$C_{14}H_{10}BrIN_2O_4$	477.04	135°C	72%
Ig			$C_{15}H_{12}BrIN_2O_4$	491.07	150°C	65%
Ih			$C_{19}H_{18}BrN_3O_3$	416.26	220°C	65%
Ii			$C_{15}H_{10}Br_2N_2O_2S$	442.12	138°C	72%
Ij			$C_{16}H_{12}BrN_3O_2S$	390.25	140°C	78%
Ik			$C_{19}H_{16}BrNO$	354.24	136	85%
Il			$C_{19}H_{13}N_3OS$	331.39	188-192°C	78%
Im			$C_{17}H_{11}ClFNO$	299.72	132 °C	70%
In			$C_{17}H_{10}Cl_2INO$	442.07	118 °C	80%
Io			$3C_{16}H_{14}BrCl_2NO_3$	419.09726	105 °C	87%
Ip			$C_{14}H_8BrCl_2NO_2$	373.02882	62 °C	80%
Iq			$C_{14}H_9IN_2O_4$	396.13673	120 °C	85%

Ir			$C_{16}H_{15}Br_2NO_2$	413.10	86°C	92%
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Spectral Discussion

IR spectra of compounds were scanned on Shimadzu FTIR spectrometer. 1H NMR spectra were recorded on a Bruker FT 200 MHz instrument using TMS as an internal standard (chemical shift are given in δ ppm) The mass spectra (MS) were recorded on an automated Finnigan MAT 1020 c mass spectrometer using ionization energy of 70 ev.

IR Spectra

IR spectra of Schiff base showed characteristics band at near region $3350-3150\text{ cm}^{-1}$ due to -OH stretching vibrations of Ph-OH. A strong sharp band in the region $1630-1615\text{ cm}^{-1}$ is observed due to C=N stretching vibration and band around $1650-1500\text{ cm}^{-1}$ due to aromatic stretching. The characteristic vibrations have been recorded at in the region of 1630 to 1430 due to Aromatic ring stretching. The presence of ether group Ph-O-C has been confirmed due to characteristic stretching vibration peaks in the band of 1310-1010.

1H NMR

1H NMR spectra of Schiff bases studied in $CDCl_3$ showed multiplet in the region δ 7.3 - 7.9 due to aromatic proton. The proton attached to the nitrile group H-C=N showed chemical shift to higher delta (δ) values ranging between 9.80 – 8.40 δ ppm. The protons of R-CH₃ group showed singlet peaks in the region 2.30 – 3.80 δ ppm.

Mass Spectra

Mass spectroscopy is important tool in determining the molecular mass of the unknown compounds. Mass spectra of Schiff base of some representative members of the series wherein good agreement with their suggested structures.

1.spectra of 4-[[*(1E)*-(3-bromo-4-hydroxy-5-methoxyphenyl) methylidene] amino}-1, 5-dimethyl-2-phenyl-1, 2-dihydro-3*H*-pyrazol-3-one (Ih)

IR: $\nu_{\max}$: 3213-3190 (Ph-OH), 2935 (C-H of R-CH₃), 2928 (C-H of R-CH₃), 1624 (C=O), 1570 (C=N), 1593, 1498, 1419 (Ar, C=C), 1380 (C-N of Ph-N-(R₂)), 1357 (C-N of (C₃)N), 1265 (Ph-O-C), 1246, 1173, 1153, 1130, 957 (Ph-C-H), 1045 (Ph-O-C), 856, 829, 764 (Ph-C-H), 705 (Aromatic ring), 586, 563 (C-Br)

^1H NMR : ^1H NMR: CDCl_3 : δ 2.49 (3H, s, CH_3), 3.16 (3H, s, CH_3), 3.90 (3H, s, CH_3), 7.37 (1H, d, Ar-H), 7.39 (1H, m, Ar-H), 7.42 (2H, m, 7.51 (1H, d, Ar-H), 7.53-7.56 (2H, m, Ar-H), 9.44 (1H, s, $\text{N}=\text{C}-\text{H}$)

MS: m/z : Mass Spectrum shows the formation of molecular ion peak (M-1) at $m/z = 415.8$ which is equal to the calculated molecular weight 416.26.

2. Spectra of 2-bromo-4- $\{(E)-[(2\text{-bromo-3,5-dimethylphenyl})\text{imino}]\text{methyl}\}$ -6- methoxyphenol (Ir)

IR: ν_{max} : 3317 (OH of Ph-OH), 3010 (Aromatic C-H), 2975 (C-H of O- CH_3), 2950 (C-H of Ph- CH_3), 2890 (C-H of O- CH_3), 2860 (C-H of Ph- CH_3), 1674 (C=N), 1606, 1599, 1504, 1461 (Aromatic ring), 1427 (C-H of O- CH_3), 1390 (OH of Ph-OH), 1354 (C H of O- CH_3), 1292 (C-O-C of Ph-O-C), 1157 (C-O of Ph-OH), 1045, 795 (C-O-C of Ph-O-C), 976 (Aromatic C-H), 679 (C-Br)

^1H NMR : ^1H NMR: CDCl_3 : δ 2.22-2.30 (3H, s, CH_3), 3.99 (6H, s, CH_3), 4.45 (3H, s, CH_3), 6.50 (1H, d, Ar-H), 7.18 (1H, d, Ar-H), 7.37 (1H, d, Ar-H), 7.64 (1H, d, Ar-H), 9.79 (1H, s, $\text{N}=\text{C}-\text{H}$)

MS: m/z : Mass Spectrum shows the formation of molecular ion peak (M+1) at $m/z = 412.9$ which is equal to the calculated molecular weight 413.1.

3. Spectra of 2-bromo-4- $\{(E)-[(2\text{-iodo-4-nitrophenyl})\text{imino}]\text{methyl}\}$ -6-methoxyphenol (If)

IR: ν_{max} : 3379 (Ph-OH), 3283 (Ar-C-H), 2927 (C-H of O- CH_3), 2846 (C-H of O- CH_3), 1666 (C=N), 1612, 1585, 1492 (Aromatic Ring), 1554 (C- NO_2), 1335 (C- NO_2), 1396 (Ph-C-O-C), 1041 (Ph-C-O-C), 1292, 1245, 1164 (Ar-C-H), 1083, 972, 856 (Ar-C H), 678 (C-I), 617 (C-Br).

^1H NMR : ^1H NMR: CDCl_3 : δ 3.99 (3H, s, OCH_3), 6.52 (1H, d, Ar-H), 6.60- 6.63 (1H, m, Ar-H), 7.37 (1H, d, Ar-H), 7.55-7.58 (1H, m, Ar-H), 7.65 (1H, m, Ar H), 9.80 (1H, s, $\text{N}=\text{C}-\text{H}$)

4. Spectra of 2-bromo-4- $\{(E)-[(5\text{-bromo-1,3-benzothiazol-2-yl})\text{imino}]\text{methyl}\}$ -6- methoxyphenol (Ii)

IR: ν_{max} : 3263 (OH of Ph-OH), 3040 (Aromatic C-H), 2955 (C-H of O- CH_3), 2733 (C-H of O- CH_3), 1674, 1630, 1582, 1444 (Aromatic Ring), 1535 (C=N), 1420 (C-H of O CH_3), 1406 (C-H of O- CH_3), 1354 (OH of Ph-OH), 1292 (C-O-C of Ph-O-C), 1276 (C-O of Ph-OH), 1210, 1160, 1047 (Aromatic C-H), 1099, 830 (C-O-C of Ph-O-C), 972, 892, 855, 765, 703 (Aromatic C-H), 564 (C-Br)

^1H NMR : ^1H NMR: CDCl_3 : δ 3.90 (3H, s, O- CH_3), 7.19 (1H, d, Ar-H), 7.28 (1H, m, Ar-H), 7.42 (1H, m, Ar-H), 7.58 (1H, m, Ar-H), 7.72 (1H, d, Ar-H), 9.77 (1H, s, $\text{H}-\text{C}=\text{N}$)

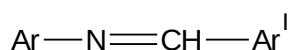
MS: m/z : Mass Spectrum shows the formation of molecular ion peak at $m/z = 440.1$ which is equal to the calculated molecular weight 442.1.

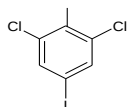
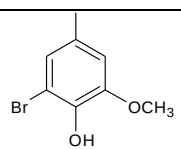
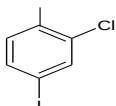
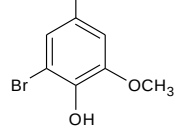
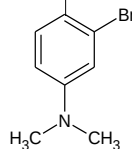
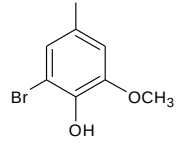
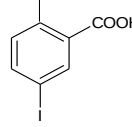
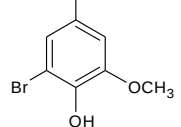
5. Spectra of 2-bromo-6-methoxy-4- $\{(E)-[(5\text{-phenyl-1,3,4-thiadiazol-2-yl})\text{imino}]\text{methyl}\}$ phenol (Ij)

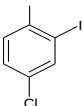
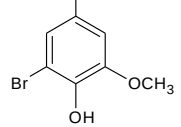
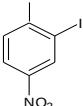
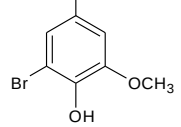
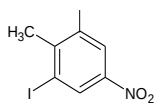
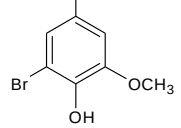
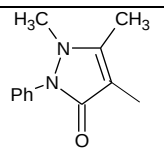
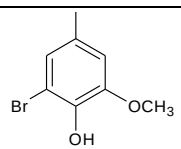
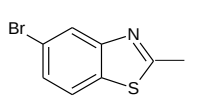
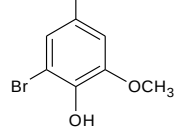
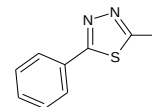
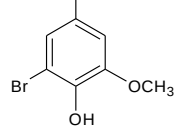
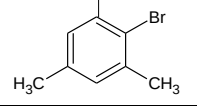
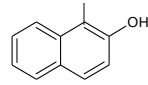
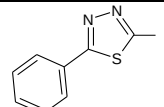
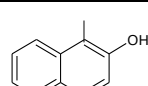
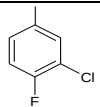
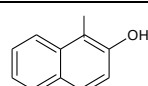
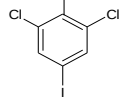
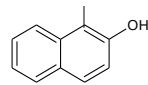
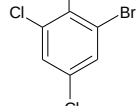
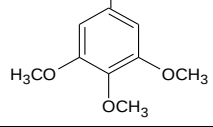
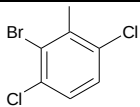
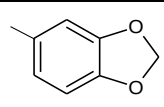
IR: $\nu_{\max}$: 2578 (OH of Ph-OH), 2348 (Aromatic C-H), 2124 (C-H of O-CH₃), 1942 (C-H of O-CH₃), 1666, 1636, 1488, 1426 (Aromatic Ring), 1583 (C=N), 1420 (C-H of O-CH₃), 1406 (C-H of O-CH₃), 1358 (OH of Ph-OH), 1295 (C-O-C of Ph-O-C), 1166 (Aromatic C-H), 1049, 830 (C-O-C of Ph-O-C), 971, 854, 794 (Aromatic C-H), 679 (Aromatic ring), 535 (C-Br)

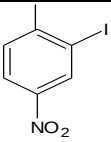
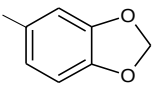
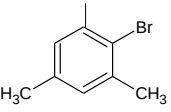
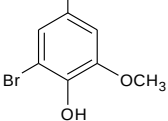
¹H NMR : ¹H NMR: DMSO: δ 3.91 (3H, s, O-CH₃), 7.41-7.50 (7H, m, Ar-C-H), 9.78 (1H, s, H-C=N)

Table 4: Antibacterial Activities of Substituted Schiff Bases I (a-r)



Compounds	Ar	Ar ^I	Zone of Inhibition in millimetre (in mm)			
			Xc	E.coli	Ec	Bs
Ia			07	08	06	02
Ib			06	07	06	03
Ic			02	03	02	00
Id			22	21	23	00

Ie			10.	12	07	00
If			12	10	11	00
Ig			06	05	06	00
Ih			02	03	02	00
Ii			20	17	22	00
Ij			02	04	02	00
Ik			07	05	07	02
Il			02	03	02	00
Im			10	12	12	00
In			12	10	14	03
Io			04	03	03	00
Ip			19	16	21	00

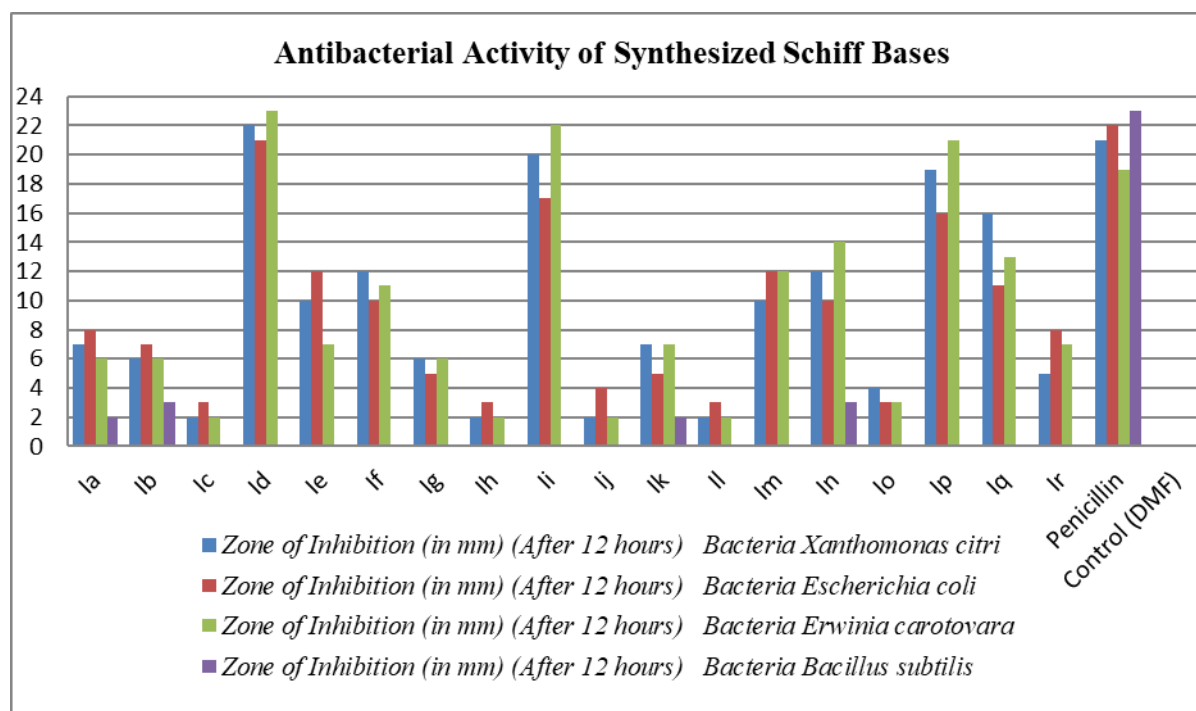
Iq			16	11	13	00
Ir			05	08	07	00
Penicillin			21	22	19	23
Control(DMF)			00	00	00	00

Xc = *Xanthomonas citri*

E. coli = *Escherichia coli*

Ec = *Erwinia carotovora*

Bs = *Bacillus subtilis*



Conclusion:

The new synthesized schiff bases were evaluated for their antibacterial activity against selected plant and human pathogenic bacterial strains. The antibacterial activity of synthesized compounds was

determined by screening them against gram negative and gram positive culture viz. *Xanthomonas citri*, *Erwinia carotovora*, *Escherichia coli*, *Bacillus subtilis* using cup plate agar diffusion method.

From the results it is clear that all the newly synthesized Schiff bases showed excellent antibacterial activity. Most remarkable activity is shown by Id, Ii, Im, In, Ip followed by the other compounds.

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